

Now, if mutual inductance is maximised by keeping the Vcc and ground tracks close together, the effective inductance will be reduced to almost half. Ideally, the Vcc track should follow the ground track on the PCB. This greatly reduces track length, and also the reduced loop area greatly reduces associated emissions and susceptibility issues.

Fig. 2(a) shows a poor PCB layout while Fig. 2(b) shows an improved

capacitor Cc is placed near each IC between power supply and ground tracks, it supplies this short-term switching current and reduces the length of tracks over which this high current flows.

The value of this capacitor (also called decoupling capacitor or decap) should ideally be around 1nF. The capacitor should be of ceramic type as these are capable of releasing their charge at whatever rate re-

quired. This and their lower self inductance makes these ideal candidates for decoupling function.

In a board consisting of a large number of ICs (more than 15), the decoupling capacitor of each IC needs to be recharged and

power bus wherein an analogue amplifier is shown sharing common power supply and ground tracks with a logic gate. The impedances of supply and ground tracks have been shown as lumped impedances (Z_g). At higher frequencies, the impedances of the tracks increase many times, not only because of increased inductance but also because of increased resistance (due to skin effect). This causes an increase in the impedance of the supply tracks.

As we have seen earlier, ground bounce occurs whenever a logic gate switches state. A part of the supply line impedance (Z_{g3}) is common to both the amplifier and the logic gate, so the amplifier will see the ground bounce as noise across power supply and the ground connection. This noise may be transferred to the amplifier circuit either directly through the power supply rails or through the common impedance Z_{g3} , and the noise voltage across Z_{g3} appears directly across the amplifier input. To reduce the common-impedance coupling we can either reduce the magnitude of the common impedance or remove it entirely.

Eliminating common impedances

The common impedances can be eliminated by using single-point or star connections between power supplies of different circuits. This can be achieved by grouping circuits depending upon their susceptibility (and emissions). Each circuit has multi-point grounding, but the groups use star connections while connecting to common power supply and ground rails. This method is called hybrid connection. The second approach is to use separate power supplies for separate circuit groups, which provides even better isolation between circuits.

This article is an extract from a book by the author. The next part will cover multi-layer PCB designing.

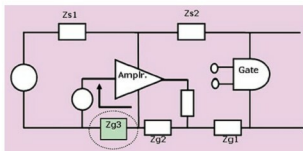


Fig. 4: Common impedance coupling

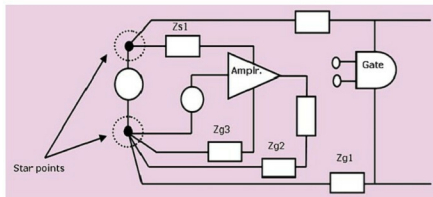


Fig. 5: Single-point grounding

PCB layout. By reducing loop area we have reduced track length and associated track impedance, and by doing so we have achieved a reduction in emissions and susceptibility.

Reservoir/decoupling capacitors

As shown in the Fig. 3(a), even if ground and Vcc tracks are placed close to each other, a high transient switching current is drawn all the way from the power supply rails, causing a large loop of high current (shown by shaded region), which can radiate. If ceramic reservoir

the recharging may occur at a considerable low frequency. A bigger bulk decoupling capacitor is often required to supply charging currents to all the decoupling capacitors. The value of this capacitor should be at least ten times the sum of values of all decoupling capacitors. All bulk decaps must have low series inductance; tantalum or metallised poly-carbonate capacitors are best suited for such applications

Common impedance coupling in PCBs

Fig. 4 shows an example of common impedance coupling in a shared